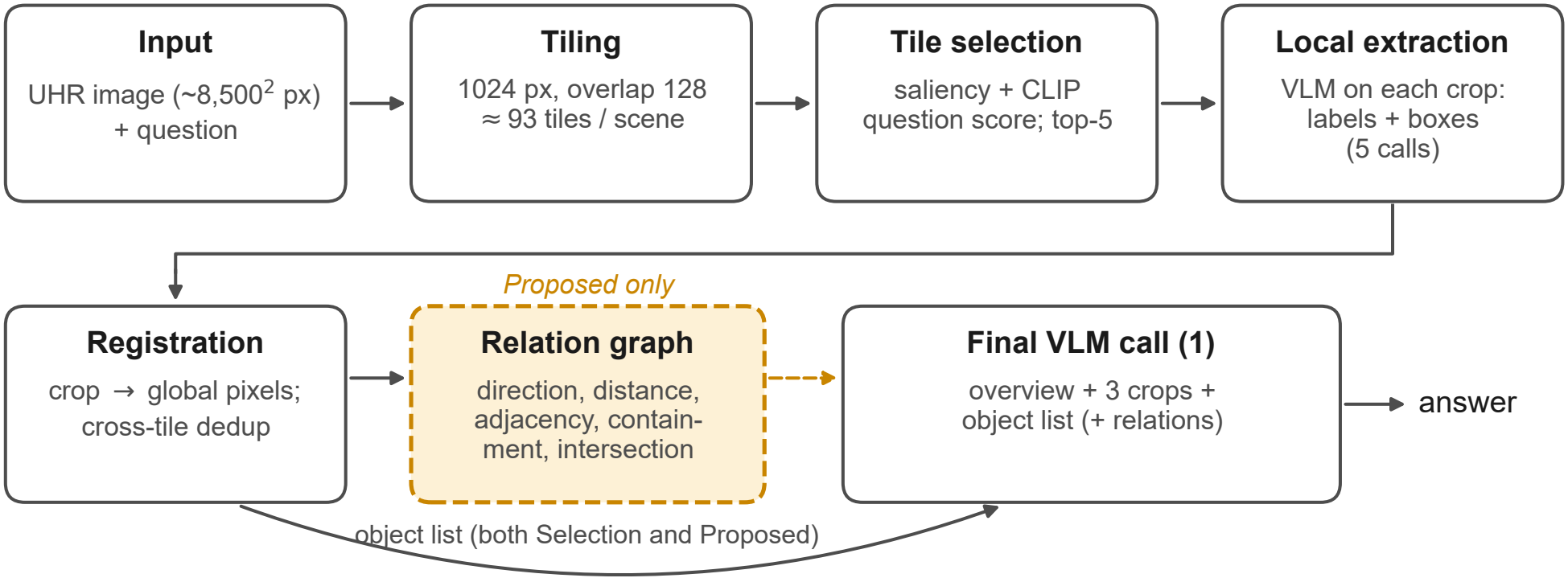

Input
UHR image (~8,500^2 px)
+ question
Tiling
1024 px, overlap 128
≈ 93 tiles / scene
Tile selection
saliency + CLIP
question score; top-5
Local extraction
VLM on each crop:
labels + boxes
(5 calls)
Registration
crop → global pixels;
cross-tile dedup
Proposed only
Relation graph
direction, distance,
adjacency, contain-
ment, intersection
Final VLM call (1)
overview + 3 crops +
object list (+ relations)
answer
object list (both Selection and Proposed)